SIM1003 Calculus II
Tutorial 6

All even-numbered questions in the following exercises

Exercise 9.7:

Intervals of Convergence

In Exercises 1–36, **(a)** find the series' radius and interval of convergence. For what values of x does the series converge **(b)** absolutely, **(c)** conditionally?

1. $\displaystyle\sum_{n=0}^{\infty} x^n$

2. $\displaystyle\sum_{n=0}^{\infty} (x+5)^n$

3. $\displaystyle\sum_{n=0}^{\infty} (-1)^n(4x+1)^n$

4. $\displaystyle\sum_{n=1}^{\infty} \frac{(3x-2)^n}{n}$

5. $\displaystyle\sum_{n=0}^{\infty} \frac{(x-2)^n}{10^n}$

6. $\displaystyle\sum_{n=0}^{\infty} (2x)^n$

7. $\displaystyle\sum_{n=0}^{\infty} \frac{nx^n}{n+2}$

8. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n(x+2)^n}{n}$

9. $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n\sqrt{n}\,3^n}$

10. $\displaystyle\sum_{n=1}^{\infty} \frac{(x-1)^n}{\sqrt{n}}$

11. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$

12. $\displaystyle\sum_{n=0}^{\infty} \frac{3^n x^n}{n!}$

13. $\displaystyle\sum_{n=1}^{\infty} \frac{4^n x^{2n}}{n}$

14. $\displaystyle\sum_{n=1}^{\infty} \frac{(x-1)^n}{n^3 3^n}$

15. $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{\sqrt{n^2+3}}$

16. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{\sqrt{n+3}}$

17. $\displaystyle\sum_{n=0}^{\infty} \frac{n(x+3)^n}{5^n}$

18. $\displaystyle\sum_{n=0}^{\infty} \frac{nx^n}{4^n(n^2+1)}$

19. $\displaystyle\sum_{n=0}^{\infty} \frac{\sqrt{n}\,x^n}{3^n}$

20. $\displaystyle\sum_{n=1}^{\infty} \sqrt[n]{n}\,(2x+5)^n$

21. $\displaystyle\sum_{n=1}^{\infty} (2+(-1)^n)\cdot(x+1)^{n-1}$

22. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n 3^{2n}(x-2)^n}{3n}$

23. $\displaystyle\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^n x^n$

24. $\displaystyle\sum_{n=1}^{\infty} (\ln n)x^n$

25. $\displaystyle\sum_{n=1}^{\infty} n^n x^n$

26. $\displaystyle\sum_{n=0}^{\infty} n!(x-4)^n$

27. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x+2)^n}{n2^n}$

28. $\displaystyle\sum_{n=0}^{\infty} (-2)^n(n+1)(x-1)^n$

29. $\displaystyle\sum_{n=2}^{\infty} \frac{x^n}{n(\ln n)^2}$

Get the information you need about $\sum 1/(n(\ln n)^2)$ from Section 9.3, Exercise 61.

30. $\displaystyle\sum_{n=2}^{\infty} \frac{x^n}{n\ln n}$

Get the information you need about $\sum 1/(n \ln n)$ from Section 9.3, Exercise 60.

31. $\displaystyle\sum_{n=1}^{\infty} \frac{(4x-5)^{2n+1}}{n^{3/2}}$

32. $\displaystyle\sum_{n=1}^{\infty} \frac{(3x+1)^{n+1}}{2n+2}$

33. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2\cdot4\cdot6\cdots(2n)}x^n$

34. $\displaystyle\sum_{n=1}^{\infty} \frac{3\cdot5\cdot7\cdots(2n+1)}{n^2\cdot2^n}x^{n+1}$

35. $\displaystyle\sum_{n=1}^{\infty} \frac{1 + 2 + 3 + \cdots + n}{1^2 + 2^2 + 3^2 + \cdots + n^2} x^n$

36. $\displaystyle\sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n}\right)(x - 3)^n$

In Exercises 37–40, find the series' radius of convergence.

37. $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{3 \cdot 6 \cdot 9 \cdots 3n} x^n$

38. $\displaystyle\sum_{n=1}^{\infty} \left(\frac{2 \cdot 4 \cdot 6 \cdots (2n)}{2 \cdot 5 \cdot 8 \cdots (3n - 1)}\right)^2 x^n$

39. $\displaystyle\sum_{n=1}^{\infty} \frac{(n!)^2}{2^n (2n)!} x^n$

40. $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2} x^n$

(*Hint:* Apply the Root Test.)

In Exercises 41–48, use Theorem 20 to find the series' interval of convergence and, within this interval, the sum of the series as a function of x.

41. $\displaystyle\sum_{n=0}^{\infty} 3^n x^n$

42. $\displaystyle\sum_{n=0}^{\infty} (e^x - 4)^n$

43. $\displaystyle\sum_{n=0}^{\infty} \frac{(x - 1)^{2n}}{4^n}$

44. $\displaystyle\sum_{n=0}^{\infty} \frac{(x + 1)^{2n}}{9^n}$

45. $\displaystyle\sum_{n=0}^{\infty} \left(\frac{\sqrt{x}}{2} - 1\right)^n$

46. $\displaystyle\sum_{n=0}^{\infty} (\ln x)^n$

47. $\displaystyle\sum_{n=0}^{\infty} \left(\frac{x^2 + 1}{3}\right)^n$

48. $\displaystyle\sum_{n=0}^{\infty} \left(\frac{x^2 - 1}{2}\right)^n$

Exercise 9.8

Finding Taylor Polynomials

In Exercises 1–10, find the Taylor polynomials of orders 0, 1, 2, and 3 generated by f at a.

1. $f(x) = e^{2x}, \quad a = 0$

2. $f(x) = \sin x, \quad a = 0$

3. $f(x) = \ln x, \quad a = 1$

4. $f(x) = \ln(1 + x), \quad a = 0$

5. $f(x) = 1/x, \quad a = 2$

6. $f(x) = 1/(x + 2), \quad a = 0$

7. $f(x) = \sin x, \quad a = \pi/4$

8. $f(x) = \tan x, \quad a = \pi/4$

9. $f(x) = \sqrt{x}, \quad a = 4$

10. $f(x) = \sqrt{1 - x}, \quad a = 0$

Finding Taylor Series at $x = 0$ (Maclaurin Series)

Find the Maclaurin series for the functions in Exercises 11–24.

11. e^{-x}

12. xe^{x}

13. $\dfrac{1}{1 + x}$

14. $\dfrac{2 + x}{1 - x}$

15. $\sin 3x$

16. $\sin \dfrac{x}{2}$

17. $7 \cos(-x)$

18. $5 \cos \pi x$

19. $\cosh x = \dfrac{e^{x} + e^{-x}}{2}$

20. $\sinh x = \dfrac{e^{x} - e^{-x}}{2}$

21. $x^4 - 2x^3 - 5x + 4$

22. $\dfrac{x^2}{x + 1}$

23. $x \sin x$

24. $(x + 1) \ln(x + 1)$

Finding Taylor and Maclaurin Series

In Exercises 25–34, find the Taylor series generated by f at $x = a$.

25. $f(x) = x^3 - 2x + 4, \quad a = 2$

26. $f(x) = 2x^3 + x^2 + 3x - 8, \quad a = 1$

27. $f(x) = x^4 + x^2 + 1, \quad a = -2$

28. $f(x) = 3x^5 - x^4 + 2x^3 + x^2 - 2, \quad a = -1$

29. $f(x) = 1/x^2, \quad a = 1$

30. $f(x) = 1/(1 - x)^3, \quad a = 0$

31. $f(x) = e^x, \quad a = 2$

32. $f(x) = 2^x, \quad a = 1$

33. $f(x) = \cos(2x + (\pi/2)), \quad a = \pi/4$

34. $f(x) = \sqrt{x + 1}, \quad a = 0$

Exercise 9.8E (Extra Exercise)

Finding Taylor Series

Use substitution to find the Taylor series at x = 0 of the functions in Exercises 1 - 12.

1. e^{-5x}

2. $e^{-x/2}$

3. $5\sin(-x)$

4. $\sin\left(\frac{\pi x}{2}\right)$

5. $\cos 5x^2$

6. $\cos\left(x^{2/3}/\sqrt{2}\right)$

7. $\ln(1+x^2)$

8. $\tan^{-1}(3x^4)$

9. $\dfrac{1}{1+\frac{3}{4}x^3}$

10. $\dfrac{1}{2-x}$

11. $\ln(3+6x)$

12. $e^{-x^2+\ln 5}$

Use power series operations to find the Taylor series at $x = 0$ for the functions in Exercises 13–30.

13. xe^x

14. $x^2\sin x$

15. $\dfrac{x^2}{2}-1+\cos x$

16. $\sin x - x + \dfrac{x^3}{3!}$

17. $x\cos\pi x$

18. $x^2\cos(x^2)$

19. $\cos^2 x$ (*Hint:* $\cos^2 x = (1+\cos 2x)/2$.)

20. $\sin^2 x$

21. $\dfrac{x^2}{1-2x}$

22. $x\ln(1+2x)$

23. $\dfrac{1}{(1-x)^2}$

24. $\dfrac{2}{(1-x)^3}$

25. $x\tan^{-1}x^2$

26. $\sin x \cdot \cos x$

27. $e^x + \dfrac{1}{1+x}$

28. $\cos x - \sin x$

29. $\frac{x}{3}\ln(1+x^2)$

30. $\ln(1+x) - \ln(1-x)$

Find the first four nonzero terms in the Maclaurin series for the functions in Exercises 31–38.

31. $e^x \sin x$

32. $\dfrac{\ln(1+x)}{1-x}$

33. $(\tan^{-1} x)^2$

34. $\cos^2 x \cdot \sin x$

35. $e^{\sin x}$

36. $\sin(\tan^{-1} x)$

37. $\cos(e^x - 1)$

38. $\cos\sqrt{x} + \ln(\cos x)$